Modular Inverses, Congruences, and Modular Powers

Mingxun Zhou (mingxunz@ust.hk)

COMP2711 Discrete Mathematical Tools

1 Modular inverses and equations

Modular inverses replace division in congruences. They allow us to solve equations such as $17x \equiv 5 \pmod{43}$ and construct solutions to simultaneous congruences. Fermat's and Euler's theorems then give rules for reducing exponents.

All variables denoting numbers are integers unless stated otherwise; moduli are greater than 1. Recall

$$a \equiv b \pmod{m} \iff m \mid (a-b) \iff a = b + km \text{ for some } k \in \mathbb{Z}.$$

Addition, subtraction, and multiplication preserve congruence. Cancellation needs an extra condition: $2 \times 1 \equiv 2 \times 4 \pmod{6}$, but $1 \not\equiv 4 \pmod{6}$.

Recall that $\gcd(a, b)$ is the greatest positive common divisor of a, b, and that a, b are coprime when $\gcd(a, b) = 1$. For $a \geq 0$ and $b > 0$, Euclid's reduction gives

$$\gcd(a, b) = \gcd(b, a \bmod b).$$

MUST UNDERSTAND

Definition (Modular inverse)**.** An inverse of a modulo $m > 1$ is an integer u satisfying $au \equiv 1 \pmod{m}$. We write $u \equiv a^{-1} \pmod{m}$ when an inverse exists.

Multiplication by u undoes multiplication by a modulo m. For example, $3 \times 5 = 15 \equiv 1 \pmod{7}$, so 5 is an inverse of 3 modulo 7.

1.1 Bézout's identity and extended Euclid

Theorem 1.1 (Bézout)**.** *For integers a, m not both zero, there exist integers u, k such that*

$$au + km = \gcd(a, m).$$

The *extended Euclidean algorithm* computes the coefficients in this identity. It runs Euclid forward to find the GCD, then works backward to express it as an integer combination of the inputs.

Proof. For $a, m > 0$, run Euclid until the pair is $(g, 0)$, where $g = \gcd(a, m)$. The final identity is $1 \times g + 0 \times 0 = g$.

Suppose a step replaces (x, y) by (y, r), where $x = qy + r$, and that $sy + tr = g$ is already known. Substituting $r = x - qy$ gives

$$g = sy + t(x - qy) = tx + (s - tq)y.$$

Thus coefficients (s, t) for the smaller pair give $(t, s - tq)$ for the preceding pair. Working backward to (a, m) proves the identity and computes its coefficients.

For nonzero inputs of either sign, apply the argument to $|a|, |m|$ and adjust the coefficient signs. If one input is zero, multiply the other by 1 or -1 to obtain its absolute value, which is the GCD. □

Algorithm 1 Extended Euclidean algorithm

Input: Nonnegative integers a, b, not both zero.
Output: Integers (g, s, t) with $g = \gcd(a, b)$ and $sa + tb = g$.

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1: procedure ExtendedEuclid(a, b)
2:     if b = 0 then
3:         return (a, 1, 0)
4:     end if
5:     q ← ⌊a/b⌋
6:     r ← a mod b
7:     (g, s, t) ← ExtendedEuclid(b, r)
8:     return (g, t, s − tq)
9: end procedure
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The recursive call solves the smaller pair. Since $sb + tr = g$ and $r = a - qb$, the returned coefficients satisfy $ta + (s - tq)b = g$. The calls follow Euclid's remainder sequence; the coefficients are computed as the calls return.

Example: An inverse of 17 modulo 43. The pairs and quotients are computed from top to bottom. The coefficients are computed from bottom to top, maintaining $sa + tb = 1$.

a	b	$q = \lfloor a/b \rfloor$	s	t
43	17	2	2	-5
17	9	1	-1	2
9	8	1	1	-1
8	1	8	0	1
1	0	—	1	0

At the bottom, $1 \times 1 + 0 \times 0 = 1$, so start with $(s, t) = (1, 0)$. Using the quotient in each preceding row, the backward updates are

$$(1, 0) \xrightarrow{q=8} (0, 1) \xrightarrow{q=1} (1, -1) \xrightarrow{q=1} (-1, 2) \xrightarrow{q=2} (2, -5).$$

For example, the last update gives $(2, -1 - 2 \times 2) = (2, -5)$. The top row yields $2 \times 43 - 5 \times 17 = 1$. Reducing modulo 43 shows $17^{-1} \equiv -5 \equiv 38 \pmod{43}$.

1.2 When inverses exist

Theorem 1.2 (Inverse criterion)**.** *For $m > 1$, a has an inverse modulo m if and only if* $\gcd(a, m) = 1$. *When an inverse exists, it is unique modulo m.*

Proof. If $\gcd(a, m) = 1$, Bézout gives $sa + tm = 1$. Reducing modulo m yields $sa \equiv 1 \pmod{m}$, so s is an inverse.

Conversely, if $au \equiv 1 \pmod{m}$, then $au - km = 1$ for some integer k. Every common divisor of a, m divides the left side, so divides 1. Hence $\gcd(a, m) = 1$.

If u, v both invert a, then

$$u \equiv u(av) = (ua)v \equiv v \pmod{m}. \qquad \square$$

Example: Solving $17x \equiv 5 \pmod{43}$**.** The preceding calculation gives $17^{-1} \equiv 38 \pmod{43}$. Multiplying by this inverse yields

$$x \equiv 38 \times 5 = 190 \equiv 18 \pmod{43}.$$

Conversely, $17 \times 18 = 306 = 7 \times 43 + 5$, so every integer $x = 18 + 43k$ is a solution. These are all solutions because multiplication by the inverse is reversible.

Cancellation. If $\gcd(c, m) = 1$, multiplying $cx \equiv cy \pmod{m}$ by an inverse of c gives $x \equiv y \pmod{m}$. In particular, every nonzero remainder modulo a prime can be cancelled.

1.3 Optional: When the coefficient is not invertible

Consider $6x \equiv 8 \pmod{14}$ and $6x \equiv 7 \pmod{14}$. The second is impossible: $6x - 14k$ is even and cannot equal 7.

For the first, use divisibility to justify the reduction:

$$6x \equiv 8 \pmod{14} \iff 14 \mid (6x - 8) \iff 7 \mid (3x - 4) \iff 3x \equiv 4 \pmod{7}.$$

The modulus must be divided too. Since $3^{-1} \equiv 5 \pmod{7}$, we obtain $x \equiv 20 \equiv 6 \pmod{7}$. Among $0, \ldots, 13$, this gives 6 and 13, so the solutions modulo 14 are

$$x \equiv 6, 13 \pmod{14}.$$

Theorem 1.3 (Linear congruences)**.** *Let $g = \gcd(a, m)$, where $m > 1$. The congruence $ax \equiv b \pmod{m}$ is solvable exactly when $g \mid b$. If solvable, there are exactly g distinct solutions among $0, \ldots, m-1$.*

Proof. A solution gives $ax - km = b$, so $g \mid b$. Conversely, Bézout gives $sa + tm = g$; if $g \mid b$, multiplication by b/g gives a solution $x = s(b/g)$.

For the count, write $a = ga'$, $b = gb'$, $m = gm'$. The congruence is equivalent to $a'x \equiv b' \pmod{m'}$. If $m' > 1$, then $\gcd(a', m') = 1$, so this has a unique solution x_0 modulo m'. Its g distinct representatives modulo m are

$$x_0,\ x_0 + m', \ldots,\ x_0 + (g-1)m'.$$

If $m' = 1$, then $g = m$, and both a and b are divisible by m; every one of the $m = g$ remainders is a solution. □

2 The Chinese remainder theorem

A system of congruences specifies several remainder conditions on the same integer. For example,

$$\begin{aligned} x &\equiv 2 \pmod{3}, \\ x &\equiv 3 \pmod{5}. \end{aligned}$$

Among $0, \ldots, 14$, the first condition allows $2, 5, 8, 11, 14$ and the second allows $3, 8, 13$. Their shared candidate is 8.

MUST UNDERSTAND

Theorem 2.1 (Chinese remainder theorem)**.** *Let $m_1, \ldots, m_n > 1$ be pairwise coprime. For any integers $r_1, \ldots, r_n$, the system*

$$\begin{aligned} x &\equiv r_1 \pmod{m_1}, \\ &\vdots \\ x &\equiv r_n \pmod{m_n} \end{aligned}$$

has exactly one solution modulo $M = m_1 \cdots m_n$.

2.1 Construction using Bézout's identity

For two congruences, the theorem says that any prescribed remainders r_1, r_2 modulo coprime $m_1, m_2 > 1$ determine exactly one solution modulo $m_1 m_2$. An algorithm for this case extends to any number of constraints: combine the first two into one congruence modulo $m_1 m_2$, combine that result with the third, then the fourth, and continue. Pairwise coprimality ensures each accumulated product is coprime to the next modulus.

For two constraints, $\gcd(m_1, m_2) = 1$, so Bézout gives integers u_1, u_2 such that

$$u_1 m_1 + u_2 m_2 = 1.$$

Multiply this identity by r_2:

$$r_2u_1m_1 + r_2u_2m_2 = r_2.$$

Modulo m_2, the second term vanishes, giving $r_2u_1m_1 \equiv r_2 \pmod{m_2}$. The first term is also a multiple of m_1, so it is zero modulo m_1. Similarly, multiplying the identity by r_1 gives the other contribution:

Term	Modulo m_1	Modulo m_2
$r_2u_1m_1$	0	r_2
$r_1u_2m_2$	r_1	0

Add the two contributions:

$$S = r_2u_1m_1 + r_1u_2m_2.$$

Modulo m_1 this is $r_1 \times 1 + r_2 \times 0 = r_1$; modulo m_2 it is $r_1 \times 0 + r_2 \times 1 = r_2$. This proves existence.

Lemma 2.2 (Coprime divisors)**.** *If a, b are coprime and $a \mid x$, $b \mid x$, then $ab \mid x$.*

Proof. By Bézout, choose integers s, t with $sa + tb = 1$. Write $x = ah = bk$ for integers h, k. Multiplying the identity by x gives

$$x = sax + tbx = sa(bk) + tb(ah) = ab(sk + th).$$

Thus $ab \mid x$. □

For uniqueness, suppose x, y are any two solutions. Subtracting their congruences gives $m_1 \mid (x - y)$ and $m_2 \mid (x - y)$. Since $\gcd(m_1, m_2) = 1$, the lemma gives $m_1m_2 \mid (x - y)$, or $x \equiv y \pmod{m_1m_2}$. All solutions are therefore $x = x_0 + km_1m_2$, where $x_0 = S \bmod (m_1m_2)$ and $k \in \mathbb{Z}$.

Example. For $x \equiv 2 \pmod 3$ and $x \equiv 3 \pmod 5$, Bézout gives

$$2 \times 3 + (-1) \times 5 = 1.$$

Thus $u_1 = 2$, $u_2 = -1$. The raw terms $3 \times 2 \times 3 = 18$ and $2 \times (-1) \times 5 = -10$ have remainders $(0, 3)$ and $(2, 0)$. Consequently

$$S = 18 + (-10) = 8.$$

All solutions are $x = 8 + 15k$, $k \in \mathbb{Z}$.

2.2 Optional: The general construction

Instead of combining congruences one at a time, we can construct a solution directly:

1. Set $M = \prod_{i=1}^{n} m_i$ and $M_i = M/m_i$.
2. Use extended Euclid to find u_i with $M_i u_i \equiv 1 \pmod{m_i}$.
3. Return $x_0 = (\sum_{i=1}^{n} r_i M_i u_i) \bmod M$.

All integer solutions are $x = x_0 + kM$, $k \in \mathbb{Z}$.

Proof. Pairwise coprimality implies $\gcd(M_i, m_i) = 1$, so each inverse exists. For a fixed j, every M_i with $i \neq j$ contains the factor m_j. Hence

$$\sum_{i=1}^{n} r_i M_i u_i \equiv r_j M_j u_j \equiv r_j \pmod{m_j}.$$

For uniqueness, every modulus divides the difference of two solutions. Repeatedly applying the two-modulus divisibility argument above shows their product divides that difference. Adding a multiple of M preserves all conditions. □

Example: Sunzi's remainder problem. Find all integers satisfying

$$\begin{aligned} x &\equiv 2 \pmod{3}, \\ x &\equiv 3 \pmod{5}, \\ x &\equiv 2 \pmod{7}. \end{aligned}$$

The direct construction uses $M = 105$:

m_i	r_i	M_i	u_i	$r_i M_i u_i$
3	2	35	2	140
5	3	21	1	63
7	2	15	1	30

Thus $S = 140 + 63 + 30 = 233 \equiv 23 \pmod{105}$, and all solutions are $x = 23 + 105k$, $k \in \mathbb{Z}$. The remainders of 23 are indeed $2, 3, 2$.

For comparison, combine the first two constraints to get $x \equiv 8 \pmod{15}$. Combining this with $x \equiv 2 \pmod{7}$ by the two-constraint construction gives

$$x \equiv 8 \times (-14) + 2 \times 15 = -82 \equiv 23 \pmod{105},$$

using Bézout's identity $1 \times 15 + (-2) \times 7 = 1$. Equivalently, substitution gives $x = 8 + 15t$, then $t \equiv 1 \pmod{7}$, and hence the same family $x = 23 + 105k$.

3 Modular powers: Fermat and Euler

Repeated squaring computes modular powers for any modulus. Fermat's and Euler's theorems can shorten the calculation by reducing the exponent when the base and modulus are coprime.

MUST UNDERSTAND

Theorem 3.1 (Fermat's little theorem)**.** *If p is prime and $1 \le a \le p-1$, then $a^{p-1} \equiv 1 \pmod{p}$.*

For any integer a not divisible by p, reduce a to its representative in $\{1, \dots, p-1\}$ to obtain the same conclusion. For every integer a, the equivalent all-integer form is $a^p \equiv a \pmod{p}$: multiply by a when $p \nmid a$, and observe that both sides are zero modulo p otherwise.

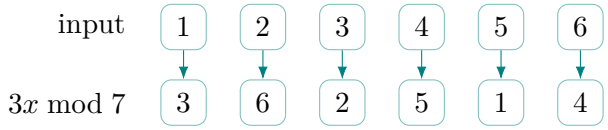
3.1 Proof of Fermat's little theorem

Let $S = \{1, 2, \dots, p-1\}$ and define $f : S \to S$ by $f(x) = ax \bmod p$. We claim that f is a bijection. Since $\gcd(a, p) = 1$, a has an inverse modulo p. If $ax \equiv 0 \pmod{p}$, cancellation gives $x \equiv 0 \pmod{p}$, impossible for $x \in S$. Thus $f(x) \in S$. If $f(x) = f(y)$, then $ax \equiv ay \pmod{p}$, so $x \equiv y \pmod{p}$. Because $x, y \in S$, this implies $x = y$. Hence f is injective. The following lemma shows that f is bijective.

Lemma 3.2 (Finite self-maps)**.** *An injective function from a finite set to itself is bijective.*

Proof. An injective map $g : T \to T$ has $|T|$ distinct outputs. Since all outputs lie in the finite set T, every element of T occurs. Thus g is surjective as well. □

Multiplying the nonzero remainders modulo 7 by 3 gives:



The outputs are $1, \dots, 6$ in a different order. In general, bijectivity gives the exact product identity

$$1 \times 2 \times \cdots \times (p-1) = f(1)f(2)\cdots f(p-1).$$

Substituting $f(i) \equiv ai \pmod{p}$ gives

$$(a \times 1)(a \times 2)\cdots(a \times (p-1)) \equiv 1 \times 2 \times \cdots \times (p-1) \pmod{p},$$

or $a^{p-1}(p-1)! \equiv (p-1)! \pmod{p}$. Every factor in $(p-1)!$ is invertible modulo p, so their product is invertible. Cancelling it proves Fermat's theorem.

Corollary 3.3 (An inverse from Fermat)**.** *If p is prime and $1 \le a \le p-1$, then*

$$a^{p-2} \equiv a^{-1} \pmod{p}.$$

Proof. $a \times a^{p-2} = a^{p-1} \equiv 1 \pmod{p}$. □

3.2 Euler's totient and theorem

Definition (Euler's totient)**.** For $m > 1$, Euler's totient $\varphi(m)$ counts the integers $x \in \{1, \dots, m-1\}$ with $\gcd(x, m) = 1$. Equivalently, it counts the remainders that have an inverse modulo m.

Modulo 15 these are $1, 2, 4, 7, 8, 11, 13, 14$, so $\varphi(15) = 8$. For a prime p, all $p-1$ nonzero remainders are invertible, giving $\varphi(p) = p - 1$.

Theorem 3.4 (Euler)**.** *If $m > 1$ and $\gcd(a, m) = 1$, then $a^{\varphi(m)} \equiv 1 \pmod{m}$. Consequently, for every $k \ge 0$,*

$$a^k \equiv a^{\,k \bmod \varphi(m)} \pmod{m}.$$

Proof. Let $r_1, \dots, r_{\varphi(m)}$ be the invertible remainders. Multiplication by a maps each to another invertible remainder, since a product of invertible numbers has an inverse. The map is injective because a can be cancelled, so it permutes this finite set. Multiplying all terms gives

$$a^{\varphi(m)} \prod_i r_i \equiv \prod_i r_i \pmod{m}.$$

The product is invertible; cancel it to get the theorem. Finally, write $k = q\varphi(m) + r$ with $0 \le r < \varphi(m)$ and replace $(a^{\varphi(m)})^q$ by 1 modulo m. □

Example. Since $\gcd(2, 15) = 1$ and $\varphi(15) = 8$, $2^8 \equiv 1 \pmod{15}$. As $100 = 12 \times 8 + 4$,

$$2^{100} \equiv 2^4 = 16 \equiv 1 \pmod{15}.$$

The coprimality condition matters: Euler's theorem cannot be applied to base 3 modulo 15; indeed $3^8 \equiv 6 \not\equiv 1 \pmod{15}$.

3.3 Computing Euler's totient

We use the convention $\varphi(1) = 1$.

Fact 1 (prime powers). For a prime p and integer $k \ge 1$,

$$\varphi(p^k) = p^k - p^{k-1} = (p-1)p^{k-1}.$$

Among $0, \dots, p^k - 1$, exactly the p^{k-1} multiples of p are not coprime to p^k. Subtracting them proves the formula. The case $k = 0$ is instead $\varphi(1) = 1$.

Fact 2 (coprime factors). For positive coprime integers a, b,

$$\varphi(ab) = \varphi(a)\varphi(b).$$

If either factor is 1, this follows from $\varphi(1) = 1$. Otherwise, CRT gives a bijection between residues modulo ab and pairs of residues modulo a, b. A residue is coprime to ab exactly when its two components are coprime to a and b, respectively. There are $\varphi(a)\varphi(b)$ such pairs, proving the identity.

For $x > 1$, write its prime factorization as $x = \prod_{i=1}^{t} p_i^{e_i}$, with distinct primes p_i and positive exponents e_i. Its prime-power factors are pairwise coprime, so the two facts give

$$\varphi(x) = \prod_{i=1}^{t} \varphi(p_i^{e_i}) = \prod_{i=1}^{t} (p_i - 1) p_i^{e_i - 1} = x \prod_{i=1}^{t} \left(1 - \frac{1}{p_i}\right).$$

For example, $60 = 2^2 \times 3 \times 5$, so

$$\varphi(60) = 60 \left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{5}\right) = 16.$$

Two distinct primes. If $p \neq q$ are prime, multiplicativity gives

$$\varphi(pq) = \varphi(p)\varphi(q) = (p - 1)(q - 1).$$

For example, $\varphi(15) = \varphi(3)\varphi(5) = 2 \times 4 = 8$.

Summary

Extended Euclid computes Bézout coefficients and modular inverses. An inverse exists exactly when the number and modulus are coprime. CRT combines congruences with pairwise-coprime moduli. Fermat's and Euler's theorems reduce modular powers; prime factorization gives a formula for Euler's totient.

4 Exercises

1. Find $12^{-1} \pmod{35}$, then solve $12x \equiv 5 \pmod{35}$.
2. Solve $15x \equiv 9 \pmod{24}$. Does $15x \equiv 10 \pmod{24}$ have a solution?
3. Find all integers satisfying $x \equiv 2 \pmod{3}$ and $x \equiv 4 \pmod{5}$.
4. Compute $3^{100} \bmod 7$ using Fermat's little theorem.
5. Compute $\varphi(60)$ from the prime factorization of 60.
6. Find $2^{100} \bmod 35$ in two ways: using Fermat and CRT, and using Euler's theorem.

Solutions

1. Since $35 = 2 \times 12 + 11$ and $12 = 11 + 1$,

$$1 = 12 - (35 - 2 \times 12) = 3 \times 12 - 35.$$

   The inverse is 3, so $x \equiv 3 \times 5 = 15 \pmod{35}$.

2. $\gcd(15, 24) = 3$. Dividing the first congruence and its modulus by 3 gives $5x \equiv 3 \pmod{8}$. Since $5^{-1} \equiv 5 \pmod{8}$, we obtain $x \equiv 7 \pmod{8}$, or $x \equiv 7, 15, 23 \pmod{24}$. The second congruence has no solution because $3 \nmid 10$.

3. The identity $2 \times 3 - 5 = 1$ gives $x \equiv 2(-5) + 4(6) = 14 \pmod{15}$. All solutions are $x = 14 + 15k$, $k \in \mathbb{Z}$.

4. Fermat gives $3^6 \equiv 1 \pmod{7}$. Since $100 = 16 \times 6 + 4$, $3^{100} \equiv 3^4 = 81 \equiv 4 \pmod{7}$.

5. $60 = 2^2 \times 3 \times 5$, so $\varphi(60) = (2 - 1)2(3 - 1)(5 - 1) = 16$.

6. Modulo 5, $2^{100} \equiv 1$; modulo 7, $2^{100} \equiv 2^4 \equiv 2$. Write $x = 1 + 5t$. Then $5t \equiv 1 \pmod{7}$, giving $t \equiv 3 \pmod{7}$ and $x \equiv 16 \pmod{35}$. Alternatively, $\varphi(35) = 24$, so Euler gives $2^{100} \equiv 2^4 = 16 \pmod{35}$.

Reference: Kenneth H. Rosen, *Discrete Mathematics and Its Applications*, 8th edition, Sections 4.3–4.4; Euler's theorem and totient calculations prepare for cryptography.